

Simultaneous primitive roots: a conditional lower bound for prime bases

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Abstract

For every fixed finite set of distinct positive primes, we prove that at least $cx/(\log x)^2$ primes in $(x, 2x)$ have every member of the set as a primitive root, for some $c > 0$ and all sufficiently large x . The result assumes four explicitly stated analytic and sieve inputs from the companion paper on primitive roots for admissible integer bases.

1 Introduction

For a prime p and an integer q not divisible by p , let $\text{ord}_p(q)$ be the multiplicative order of q in $\mathbb{F}_p^\times$. Thus q is a primitive root modulo p when $\text{ord}_p(q) = p - 1$. We study primes for which every member of a fixed finite collection of positive prime bases is a primitive root.

The problem for one base is a case of Artin’s primitive-root conjecture, recorded by Hasse in 1927 [4, Section 4.8]. Hooley proved the predicted asymptotic for one fixed integer under suitable generalized Riemann hypotheses [7]. Matthews obtained density results for simultaneous primitive roots under generalized Riemann hypotheses [8]. Anwar and Pappalardi later established an infinitude criterion under Schinzel’s Hypothesis H [1]. Unconditional results allowing a finite choice of bases were obtained by Gupta and Murty [3] and by Heath-Brown [6]. In particular, Heath-Brown’s theorem implies that at least one member of any triple of distinct positive primes is a primitive root modulo infinitely many primes.

Our result is an implication from four statements taken from the companion manuscript [P]: a uniform Hecke zero-free assertion (Input 3.1), a marked Type II estimate (Input 4.1), a block sieve (Input 5.1), and a rough-number density estimate (Input 5.2). We reproduce these statements in the precise forms used here. They are hypotheses of the following theorem; their proofs in [P] and its cited companion manuscripts are outside the scope of this paper.

Theorem 1.1. *Assume Inputs 3.1, 4.1, 5.1, and 5.2. Let $k \geq 1$ and let $q_1, \dots, q_k$ be distinct positive primes. There are constants $c_{\mathbf{q}} > 0$ and $x_{\mathbf{q}} > 1$ such that, for every real $x \geq x_{\mathbf{q}}$, at least*

$$c_{\mathbf{q}} \frac{x}{(\log x)^2}$$

primes $p \in (x, 2x)$ satisfy

$$p \nmid q_1 \cdots q_k, \quad \text{ord}_p(q_i) = p - 1 \quad (1 \leq i \leq k).$$

We extend the construction of primes with controlled predecessors in [P, Section 12] to a fixed progression. The marked Type II estimate remains available after multiplication by

a fixed Dirichlet character, and Bombieri–Vinogradov supplies the required distribution in that progression. A direct sieve for the final balanced semiprimes keeps its constant independent of the later marking parameters. The progression makes every prescribed base a quadratic nonresidue. Uniform splitting estimates in the individual Kummer fields then remove the remaining prime divisors of each order index. Section 2 gives this reduction; the subsequent sections prove the two estimates it uses.

2 Reduction to two estimates

Write $L = \log x$ and set $\eta = 10^{-6}$. For a prime $p \nmid q$, write

$$\iota_p(q) = \frac{p-1}{\text{ord}_p(q)}$$

for the order index of q modulo p . The first proposition supplies primes whose predecessors have one large prime factor and otherwise only prime factors in a controlled interval.

Proposition 2.1 (A prime reservoir in a fixed progression). *Assume Inputs 4.1, 5.1, and 5.2. Let $M \geq 1$ be odd and squarefree, and let a be a residue class modulo M with $\gcd(a(a-1), M) = 1$. There are constants $c_{M,a} > 0$ and $x_{M,a} > 1$ such that, for every real $x \geq x_{M,a}$, at least $c_{M,a}x/L^2$ primes p satisfy*

$$x < p < 2x, \quad p \equiv a \pmod{M}, \quad p = 4rQ + 1,$$

where r is a positive integer, $Q > x^{0.9}$ is prime, and

$$\exp(L^{0.1}) < \ell < \exp(L^{0.3}) \quad \text{for every prime } \ell \mid r.$$

The case $M = 1$ is included.

The second proposition bounds the exceptional primes at which a controlled prime factor can divide the index of a prescribed base.

Proposition 2.2 (A uniform bound for order obstructions). *Assume Input 3.1. For each fixed positive prime q , for all sufficiently large real x , and uniformly over primes ℓ with $\exp(L^{0.1}) < \ell < \exp(L^{0.3})$, one has*

$$\# \left\{ p \text{ prime} : \begin{array}{l} x < p < 2x, \quad p \nmid q\ell, \quad p \equiv 1 \pmod{\ell}, \\ q^{(p-1)/\ell} \equiv 1 \pmod{p} \end{array} \right\} \ll_q \frac{x}{\ell(\ell-1)L} + x^{1-\eta}.$$

Proof of Theorem 1.1. Let M be the product of the odd members of $\{q_1, \dots, q_k\}$, with $M = 1$ if this set has no odd member. For each odd q_i , choose a nonzero quadratic nonresidue modulo q_i . The Chinese remainder theorem supplies a class $a \pmod{M}$ having all these prescribed residues; when $M = 1$, take its unique residue class. Since 1 is a square, $\gcd(a(a-1), M) = 1$.

Apply Proposition 2.1, and call the resulting set of primes $\mathcal{R}(x)$. For large x , all its primes exceed the q_i , and both r and Q in $p = 4rQ + 1$ are odd. Thus $p \equiv 5 \pmod{8}$, in particular $p \equiv 1 \pmod{4}$. Quadratic reciprocity and the supplementary law for 2 give

$$\left(\frac{q_i}{p}\right) = \begin{cases} \left(\frac{p}{q_i}\right) = \left(\frac{a}{q_i}\right) = -1, & q_i > 2, \\ (-1)^{(p^2-1)/8} = -1, & q_i = 2. \end{cases}$$

For a fixed one of the bases $q = q_i$, the index $\iota_p(q)$ is odd: if it were even, then $\text{ord}_p(q) \mid (p-1)/2$, contrary to Euler's criterion. If $\iota_p(q) > 1$, it has an odd prime divisor λ . Since $\iota_p(q) \mid 4rQ$, either $\lambda = Q$ or $\lambda \mid r$, and in either case

$$q^{(p-1)/\lambda} \equiv 1 \pmod{p}. \quad (2.1)$$

If $Q \mid \iota_p(q)$, the integer $j = (p-1)/Q$ is less than $2x^{0.1}$, and $p \mid q^j - 1$. The product of $q^j - 1$ over $1 \leq j \leq 2x^{0.1}$ has logarithm at most

$$(\log q) \sum_{j \leq 2x^{0.1}} j \ll_q x^{0.2}.$$

Each of its prime divisors in $(x, 2x)$ contributes more than L to this logarithm. Consequently the number of these exceptional primes is

$$O_q\left(\frac{x^{0.2}}{L}\right) = o\left(\frac{x}{L^2}\right). \quad (2.2)$$

In the other case, write $\ell = \lambda \mid r$. The controlled interval in Proposition 2.1 applies to ℓ . Since $\ell \mid p-1$, one has $p \equiv 1 \pmod{\ell}$, while Equation (2.1) supplies the other congruence in Proposition 2.2. Sum that proposition over all such primes ℓ . The sum of $1/(\ell(\ell-1))$ is at most the telescoping sum over all integers exceeding $\exp(L^{0.1})$, and there are at most $\exp(L^{0.3})$ possible ℓ . The number of exceptions of this second kind is therefore

$$O_q\left(\frac{x}{L} \exp(-L^{0.1}) + \exp(L^{0.3}) x^{1-\eta}\right) = o\left(\frac{x}{L^2}\right). \quad (2.3)$$

Here the two ratios to x/L^2 are bounded by constant multiples of $L \exp(-L^{0.1})$ and $L^2 \exp(L^{0.3} - \eta L)$, respectively, and both tend to zero.

There are only finitely many bases. The union over them of the exceptions in Equations (2.2) and (2.3) is still $o(x/L^2)$. Since $\#\mathcal{R}(x) \geq c_{M,a}x/L^2$, for every sufficiently large real x at least $c_{M,a}x/(2L^2)$ primes in $\mathcal{R}(x)$ have $\iota_p(q_i) = 1$ for every i . These primes give the theorem. $\square$

3 A uniform splitting estimate

The analytic input is the following assertion of [P, Theorem 1.2]. A finite-order Hecke character of a number field F means a continuous finite-image character of $F^\times \backslash \mathbb{A}_F^\times$.

Input 3.1 (Uniform Hecke zero-free assertion). Let F be a cyclotomic number field containing μ_{12} . For every finite-order Hecke character χ of F , the meromorphic continuation of $L_F(s, \chi)$ has no zeros in

$$\Re s > 1 - 10^{-6}.$$

The assertion permits the pole at $s = 1$ for the principal character. We write $\eta = 10^{-6}$ throughout this section.

Proof of Proposition 2.2. Fix the positive prime q , put $L = \log x$, and let ℓ be a prime in the range of the proposition. We take x large enough in terms of q that $\ell > q$ and $\ell \geq 5$. Set

$$E_\ell = \mathbb{Q}(\mu_\ell), \quad K = K_{\ell,q} = E_\ell(q^{1/\ell}), \quad n = [K : \mathbb{Q}], \quad D = |d_K|.$$

The field K is the splitting field of $T^\ell - q$, so it is Galois over $\mathbb{Q}$. The cyclotomic discriminant is $|d_{E_\ell}| = \ell^{\ell-2}$. Since $q \neq \ell$, the prime q is unramified in E_ℓ , and its valuation at each prime of E_ℓ above q is one. The polynomial $T^\ell - q$ is Eisenstein at such a prime. Consequently

$$[K : E_\ell] = \ell, \quad n = \ell(\ell - 1). \quad (3.1)$$

The discriminant of the power basis generated by $q^{1/\ell}$ is, up to sign, $\ell^\ell q^{\ell-1}$. The relative field discriminant therefore divides $(\ell^\ell q^{\ell-1})\mathcal{O}_{E_\ell}$. The discriminant tower formula gives

$$D \leq \ell^{\ell(\ell-2)} (\ell^\ell q^{\ell-1})^{\ell-1} = \ell^{\ell(2\ell-3)} q^{(\ell-1)^2}, \quad (3.2)$$

$$\log D + n \ll_q n \log(2\ell).$$

The same divisibility and the cyclotomic discriminant show that rational primes outside $q\ell$ are unramified in K .

We identify the two congruences in Proposition 2.2 with complete splitting. Let $p \nmid q\ell$ be a prime, choose a prime $\mathfrak{P}$ of K above p , and write $\sigma = \text{Frob}_{\mathfrak{P}}$ for arithmetic Frobenius. Its restriction to E_ℓ sends ζ_ℓ to ζ_ℓ^p , so it is trivial exactly when $p \equiv 1 \pmod{\ell}$. Under this condition, put $\alpha = q^{1/\ell}$. Since $p \nmid q$, α is a unit at $\mathfrak{P}$, and the defining congruence for arithmetic Frobenius gives

$$\frac{\sigma(\alpha)}{\alpha} \in \mu_\ell, \quad \frac{\sigma(\alpha)}{\alpha} \equiv \alpha^{p-1} = q^{(p-1)/\ell} \pmod{\mathfrak{P}}.$$

Reduction on μ_ℓ is injective at $p \neq \ell$. Hence $q^{(p-1)/\ell} \equiv 1 \pmod{p}$ is equivalent to σ fixing α . Together the two congruences say that σ fixes both E_ℓ and α , and hence is the identity on K . Because $K/\mathbb{Q}$ is Galois, this is precisely complete splitting of p .

We transfer Input 3.1 to ζ_K . Put

$$F_\ell = \mathbb{Q}(\mu_{12\ell}), \quad \widetilde{K} = KF_\ell = F_\ell(q^{1/\ell}).$$

Because F_ℓ contains μ_ℓ , the extension $\widetilde{K}/F_\ell$ is Galois and the map $\sigma \mapsto \sigma(q^{1/\ell})/q^{1/\ell}$ embeds its Galois group in μ_ℓ . It is thus cyclic. Also $F_\ell/\mathbb{Q}$ is abelian Galois, so $\widetilde{K}/K$ is Galois and restriction embeds its Galois group in $\text{Gal}(F_\ell/\mathbb{Q})$. It too is abelian. Abelian Artin formalism [9, Chapter VII, Corollary (10.5) and Theorem (10.6)], with the one-dimensional Artin functions identified with their associated primitive Hecke functions, gives

$$\zeta_{\widetilde{K}}(s) = \prod_{\chi \in \widehat{\text{Gal}(\widetilde{K}/F_\ell)}} L_{F_\ell}(s, \chi),$$

$$\frac{\zeta_{\widetilde{K}}(s)}{\zeta_K(s)} = \prod_{\substack{\chi \in \widehat{\text{Gal}(\widetilde{K}/K)} \\ \chi \neq 1}} L_K(s, \chi). \quad (3.3)$$

These are identities of meromorphic functions, including the local factors at ramified primes. The first product is zero-free on $\Re s > 1 - \eta$ by Input 3.1, since F_ℓ is cyclotomic and contains μ_{12} ; its principal factor has the permitted pole at one. Each factor of the second product is an entire nonprincipal finite-order Hecke L -function [10]. A zero of ζ_K in this half-plane away from one would therefore be a zero of $\zeta_{\widetilde{K}}$. Hence

$$\zeta_K(s) \neq 0 \quad (\Re s > 1 - \eta, \ s \neq 1). \quad (3.4)$$

The common global half-plane in the input makes this width uniform in ℓ , without requiring uniform onsets for the auxiliary estimates in its proof. This is the abelian tower argument of [P, Section 9, Equation (9.2)], specialized to the base q .

To count completely split primes uniformly as the field varies, we use the prime-ideal calculation from [P, Section 9, Equations (9.5)–(9.9)]. Fix a nonnegative $\phi \in C_c^\infty((0, \infty))$ with $\phi \geq 1$ on $[1, 2]$, and put

$$\Phi(s) = \int_0^\infty \phi(t)t^{s-1} dt, \quad \Psi_{K'}(x) = \sum_{\mathfrak{p}} \sum_{m \geq 1} \log N\mathfrak{p} \phi\left(\frac{(N\mathfrak{p})^m}{x}\right)$$

for a number field K' . Suppose that K' has degree $n' = r_1 + 2r_2$, absolute discriminant D' , and no zeta zero in $\Re s > 1 - \eta$. Its nontrivial zeros ρ lie in $0 < \Re \rho < 1$, and in fact $\Re \rho \leq 1 - \eta$. The Mellin transform Φ is entire and has arbitrary inverse-power decay in height on fixed vertical strips. Mellin inversion of the logarithmic derivative, followed by a shift from $\Re s = 2$ to $\Re s = -1/2$, gives

$$\Psi_{K'}(x) = x\Phi(1) - \sum_{\rho} x^{\rho}\Phi(\rho) - (r_1 + r_2 - 1)\Phi(0) + O_{\phi}\left(x^{-1/2}(\log D' + n')\right). \quad (3.5)$$

Here zeros are counted with multiplicity. The term at the origin comes from the zero of $\zeta_{K'}$ there of order $r_1 + r_2 - 1$; it is absent when that order is zero. There are no other trivial zeros between the two lines. To see the uniformity of the error, the functional equation gives on $\Re s = -1/2$

$$\left| \frac{\zeta'_{K'}}{\zeta_{K'}}(s) \right| \ll \log D' + n' \log(2 + |\Im s|).$$

Indeed the Euler series at $1 - s$, with real part $3/2$, is $O(n')$, and the logarithmic derivatives of the archimedean factors contribute $O(n' \log(2 + |\Im s|))$. The rapid decay of Φ gives the error in Equation (3.5). The usual rectangular contours may be taken with horizontal edges away from zeros. The zero count below supplies such edges, and the Hadamard partial-fraction formula bounds the logarithmic derivative polynomially on them for each fixed K' . The decay of Φ then makes their integrals vanish; the same zero count makes the zero sum absolutely convergent.

For the required zero count, let $N_{K'}(H)$ count nontrivial zeros with $|\Im \rho| \leq H$, including multiplicity. The uniform formula of [5, Corollary 1.2, Equation (1.3)] is

$$N_{K'}(H) = \frac{H}{\pi} \log \left(D' \left(\frac{H}{2\pi e} \right)^{n'} \right) + O(\log D' + n' \log H + n') \quad (H \geq 1).$$

For $|T| \geq 2$, the unit band is bounded by $N_{K'}(|T| + 2) - N_{K'}(|T| - 1)$. The main term changes by $O(\log D' + n' \log(2 + |T|))$ over this interval, and both errors have the same bound. For $|T| < 2$, use $N_{K'}(3)$. Thus

$$\#\{\rho : T \leq \Im \rho < T + 1\} \ll \log D' + n' \log(2 + |T|), \quad (3.6)$$

uniformly in K' and T . Using it with the decay of Φ on $0 \leq \Re s \leq 1$ gives, for $x \geq 1$,

$$\begin{aligned} \sum_{\rho} |x^{\rho}\Phi(\rho)| &\ll_{\phi} x^{1-\eta} \sum_{j \in \mathbb{Z}} (1 + |j|)^{-3} (\log D' + n' \log(2 + |j|)) \\ &\ll_{\phi} x^{1-\eta} (\log D' + n'). \end{aligned} \quad (3.7)$$

The origin term in Equation (3.5) is nonpositive, since $\Phi(0) \geq 0$. Equations (3.5) and (3.7) thus imply the uniform upper bound

$$\Psi_{K'}(x) \ll_{\phi} x + x^{1-\eta} (\log D' + n') \quad (x \geq 1). \quad (3.8)$$

In particular, the implied constant depends only on the fixed test function, and not on the field.

Apply Equation (3.8) to $K' = K$, using Equation (3.4). Every rational prime p splitting completely in K supplies n prime ideals of norm p . When $x < p < 2x$, their first-power terms contribute at least $n \log p > nL$ to $\Psi_K(x)$. Equations (3.1) and (3.2) therefore give

$$\begin{aligned} & \#\{p \text{ prime} : x < p < 2x, p \text{ splits completely in } K\} \\ & \ll_q \frac{x}{\ell(\ell-1)L} + \frac{x^{1-\eta} \log(2\ell)}{L} \ll_q \frac{x}{\ell(\ell-1)L} + x^{1-\eta}. \end{aligned} \quad (3.9)$$

The last inequality uses $\log \ell < L^{0.3}$ and sufficiently large x . By the Frobenius identification above, Equation (3.9) proves the proposition. $\square$

4 A marked family in a fixed progression

We now prepare the weighted family used to prove Proposition 2.1. Fix an odd squarefree positive integer M and a residue $a \pmod{M}$ satisfying

$$\gcd(a(a-1), M) = 1. \quad (4.1)$$

The case $M = 1$ means the trivial progression. Throughout this section, x is real and tends to infinity, and

$$L = \log x, \quad W = \exp(L^{0.24}), \quad V(y) = \prod_{p \leq y} \left(1 - \frac{1}{p}\right) \quad (y \geq 1). \quad (4.2)$$

Products indexed by p are over primes. Write $P^-(h)$ for the least prime factor of $h > 1$, and put $P^-(1) = \infty$. An integer with $P^-(h) > W$ is called W -rough. Every parameter below, apart from an explicitly displayed dependence on x , is fixed before x grows.

For an integer $K \geq 1$ and fixed exponents $0.1 < a_1 < \dots < a_K < 0.2$, define the prime groups

$$\mathcal{P}_i = \{p \text{ prime} : \exp(L^{a_i}) \leq p \leq \exp(2L^{a_i})\}, \quad V_i = \sum_{p \in \mathcal{P}_i} \frac{1}{p}. \quad (4.3)$$

These groups are disjoint for sufficiently large x , and Mertens' theorem gives $V_i = \log 2 + o(1)$. We shall choose the sieve parameters first, then a sufficiently large K , and then any admissible fixed exponents a_i . Thresholds for x may depend on all these choices.

For a positive integer h , let

$$h_{\mathcal{P}} = \prod_{p \in \bigcup_i \mathcal{P}_i} p^{v_p(h)}, \quad \omega_i(h) = \sum_{p \in \mathcal{P}_i} \mathbf{1}_{p|h}. \quad (4.4)$$

Thus $h_{\mathcal{P}}$ includes every power of every group prime dividing h . Let $\mathcal{G}$ be the set of positive integers supported on the group primes, including 1; we call these the group integers. The marks of [P, Section 10], with mark parameter $1/2$, are

$$\mathcal{W}(h) = 2^{K - \sum_i \omega_i(h)} \prod_{i=1}^K \frac{\omega_i(h)}{V_i}. \quad (4.5)$$

They vanish unless every group supplies a prime divisor. Since $j2^{1-j} \leq 1$ for every integer $j \geq 1$, for sufficiently large x we have

$$0 \leq \mathcal{W}(h) \leq B_K, \quad B_K = \left(\frac{2}{\log 2}\right)^K, \quad h \geq 1. \quad (4.6)$$

The constant B_K does not depend on the exponents; the threshold may. A function F on the positive integers satisfies $F(ph) = F(h)$ for every group prime p and every $h \geq 1$ if and only if it depends only on $h/h_{\mathcal{P}}$. This equivalence includes the case $p \mid h$, so the invariance removes all powers of a group prime. In the application, F will select $h/h_{\mathcal{P}} = 4Q$ with $Q > x^{0.9}$ prime, so the complete group part supplies the factor r in $d - 1 = 4rQ$.

The rough-factor density used below is, for $w > \gamma > 0$,

$$D_{\gamma}(w) = \frac{1}{w} + \sum_{j \geq 2} \frac{1}{j!} \int_{\substack{t_i \geq \gamma \\ \sum_{i < j} t_i \leq w - \gamma}} \frac{1}{w - \sum_{i < j} t_i} \prod_{i < j} \frac{dt_i}{t_i}. \quad (4.7)$$

This is the normalization in [P, Section 10]. Its j -th term is zero unless $w \geq j\gamma$, so the sum is locally finite. It is nonnegative and jointly continuous on $w > \gamma > 0$. For completeness, in a term with $j \geq 2$ put $s = w - j\gamma$. When $s \geq 0$, the substitution $t_i = \gamma + sy_i$ writes that term as

$$\frac{s^{j-1}}{j!} \int_{\substack{y_i \geq 0 \\ \sum_{i < j} y_i \leq 1}} \frac{\prod_{i < j} dy_i}{\prod_{i < j} (\gamma + sy_i)(\gamma + s(1 - \sum_{i < j} y_i))}.$$

On any compact subset of $w > \gamma > 0$ the denominators stay bounded away from zero. Extending this expression by zero for $s < 0$ proves continuity also at $w = j\gamma$. Local finiteness completes the claim.

All Dirichlet characters in the following input, including principal characters, are extended by zero on nonunits. All frequencies are real. Part (i) of the next input transfers scalar character cancellation to the marked bilinear sum. Part (ii) supplies that cancellation for the difference of the two roughness tests, with $D_{\gamma}(\log m/L)/(LV(W))$ as the comparison factor. These are the coefficient criterion of [P, Lemma 10.2], specialized to the marks above, and the cancellation verified in the proof of [P, Proposition 10.3].

Input 4.1 (Marked Type II estimate). *(i) Coefficient criterion.* Fix $\delta, C, D_* > 0$. Suppose $H_m, H_n \geq x^{\delta}$ and $X = H_m H_n \asymp x$, with fixed comparison constants. Let $F : \mathbb{Z}_{>0} \rightarrow \mathbb{C}$ satisfy $|F(h)| \leq 1$ and $F(ph) = F(h)$ for every group prime p and every $h \geq 1$. Let (α_m) and (β_n) be scalar sequences supported on W -rough integers in $[H_m, 2H_m]$ and $[H_n, 2H_n]$, respectively, with α supported on an arbitrary interval $I \subseteq [H_m, 2H_m]$, and with $|\alpha_m|, |\beta_n| \leq L^C$. Suppose that for every fixed $A_0, B, A > 0$, every Dirichlet character χ modulo $k \leq L^{A_0}$, and every $|t| \leq 2XL^B$,

$$\left| \frac{1}{H_m} \sum_m \alpha_m \chi(m) m^{it} \right| \ll_A L^{-A}, \quad (4.8)$$

uniformly in the permitted interval, character, and frequency. The constant and threshold in this hypothesis may depend on A_0, B, A , the fixed setup, and all further fixed parameters defining the coefficients. Then

$$\left| \sum_{m,n} \alpha_m \beta_n F(mn-1) \mathcal{W}(mn-1) \right| \ll XL^{-D_*} \quad (4.9)$$

when K is sufficiently large in terms of the fixed data, independently of the particular a_i . Its implicit constant and threshold may depend on all fixed data, including the common constants and thresholds in (4.8) and the a_i . With those data fixed, the bound is uniform over the permitted F , dyads, intervals, and coefficients.

(ii) Rough coefficient cancellation. Retain the marked data and the hypotheses on F, β, H_m, H_n , without assuming a previously chosen α . Fix $0 < \gamma < w_- < w_+ < 1$, and assume

$$[\log H_m/L, \log(2H_m)/L] \subseteq [w_-, w_+]. \quad (4.10)$$

For every interval $I \subseteq [H_m, 2H_m]$ and every $|v| \leq L^C$, the coefficient

$$\alpha_m^{(0)} = m^{iv} \left(\mathbf{1}_{P^-(m) > x^\gamma} - \frac{D_\gamma(\log m/L)}{LV(W)} \mathbf{1}_{P^-(m) > W} \right) \quad (m \in I), \quad (4.11)$$

set equal to zero outside I , satisfies (4.8) with $\alpha^{(0)}$ in place of α , for every fixed $A_0, B, A > 0$. This assertion is uniform in the permitted dyads, interval, character, frequency t , and v . Its constant and threshold may depend on the fixed parameters, including γ, w_-, w_+ ; in particular γ and $w_- - \gamma$ are fixed before x grows.

Input 4.1(i) permits F to vary with x , since the estimate is uniform over all functions satisfying its two displayed conditions. The next lemma inserts the fixed progression into the coefficient.

Lemma 4.2 (A rough Type II estimate in a fixed progression). *Fix $\delta, C, D_* > 0$ and $0 < \gamma < w_- < w_+ < 1$. Let F, H_m, H_n, X satisfy the hypotheses of Input 4.1, including (4.10). Let χ_0 be any Dirichlet character modulo the fixed M . On an arbitrary interval $I \subseteq [H_m, 2H_m]$, put*

$$\alpha_m = \chi_0(m) m^{iv} \left(\mathbf{1}_{P^-(m) > x^\gamma} - \frac{D_\gamma(\log m/L)}{LV(W)} \mathbf{1}_{P^-(m) > W} \right), \quad |v| \leq L^C, \quad (4.12)$$

and set $\alpha_m = 0$ outside I . Let β be supported on W -rough integers in $[H_n, 2H_n]$, with $|\beta_n| \leq L^C$. Then

$$\left| \sum_{m,n} \alpha_m \beta_n F(mn-1) \mathcal{W}(mn-1) \right| \ll XL^{-D_*} \quad (4.13)$$

when K is sufficiently large in terms of the fixed data, independently of the particular a_i . The estimate is uniform in the permitted F , dyads, intervals, β , real v , and characters χ_0 . The implicit constant and threshold may depend on all fixed data, including M and the a_i .

Proof. Let $\alpha^{(0)}$ be the coefficient in (4.11). For a character χ modulo $k \leq L^{A_0}$, the product $\theta = \chi_0 \chi$, with the stated zero extensions, is a character modulo $h = \text{lcm}(M, k)$. Indeed it is a homomorphism on the units modulo h , and it vanishes exactly on the nonunits modulo h . Since M is fixed,

$$h \leq ML^{A_0} \leq L^{A_0+1}$$

for sufficiently large x . Input 4.1(ii), applied with exponent $A_0 + 1$, therefore gives

$$\left| \frac{1}{H_m} \sum_m \alpha_m \chi(m) m^{it} \right| = \left| \frac{1}{H_m} \sum_m \alpha_m^{(0)} \theta(m) m^{it} \right| \ll_A L^{-A} \quad (|t| \leq 2XL^B).$$

This includes imprimitive θ and the case where θ is principal. It is uniform in χ_0 , since all the product characters belong to the same enlarged modulus range.

These scalar sums and their permitted test ranges do not depend on K or the a_i : W is defined from x alone. Fix their common cancellation bounds using any one admissible auxiliary marked setup in Input 4.1(ii). The same scalar bounds then apply unchanged when the K and bands used in Input 4.1(i) are chosen.

For large x , $x^\gamma > W$, so both terms of α are supported on W -rough integers. The density is bounded on $[w_-, w_+]$, and Mertens' theorem gives $LV(W) \asymp L^{0.76}$. Hence α is bounded, and in particular $|\alpha_m| \leq L^C$ eventually; multiplication by χ_0 does not increase its size. All the hypotheses of Input 4.1(i) now hold, proving the lemma. $\square$

The product progression will be imposed by the exact orthogonality identity

$$\mathbf{1}_{mn \equiv a \pmod{M}} = \frac{1}{\varphi(M)} \sum_{\chi_0 \pmod{M}} \overline{\chi_0(a)} \chi_0(m) \chi_0(n). \quad (4.14)$$

If either factor is a nonunit modulo M , both sides are zero because a is a unit. For $M = 1$, the sum consists of the single trivial character. This identity leaves F with its required invariance.

We now choose the invariant function to encode the desired predecessors. Fix a smooth function Ψ compactly supported in $(1, 2)$, with $0 \leq \Psi \leq 1$ and $A_\Psi = \int_1^2 \Psi(u) du > 0$. From now on take

$$\begin{aligned} F(h) &= \mathbf{1}_{\{h/h_{\mathcal{P}}=4Q \text{ for a prime } Q > x^{0.9}\}}, & h \geq 1, \\ w(d) &= \mathbf{1}_{d \equiv a \pmod{M}} \Psi(d/x) F(d-1) \mathcal{W}(d-1), & d \geq 2. \end{aligned} \quad (4.15)$$

The quotient $h/h_{\mathcal{P}}$ is unchanged by multiplication by any group prime, even one already dividing h , so this F is permitted in Lemma 4.2. For large x , every group prime is coprime to $2M$ and lies in $(\exp(L^{0.1}), \exp(L^{0.3}))$, while every prime $Q > x^{0.9}$ lies outside the groups. Consequently, if $w(d) > 0$, there is a unique representation

$$\begin{aligned} x < d < 2x, & \quad d \equiv a \pmod{M}, & \quad d = 4rQ + 1, \\ r = (d-1)_{\mathcal{P}} \in \mathcal{G}, & \quad Q > x^{0.9} \text{ prime.} \end{aligned} \quad (4.16)$$

Moreover $\mathcal{W}(d-1) = \mathcal{W}(r)$ and $r < (2x)/(4x^{0.9}) \leq x^{0.11}$. We also have $0 \leq w(d) \leq B_K$.

For $r \in \mathcal{G}$, the progression in (4.15) prescribes the reduced class

$$b_r \equiv (a-1)(4r)^{-1} \pmod{M}$$

for Q . It is reduced by (4.1); inverses and classes for $M = 1$ have their trivial meaning. Define

$$\begin{aligned} A_r &= \sum_{\substack{Q > x^{0.9} \text{ prime} \\ Q \equiv b_r \pmod{M}}} \Psi((4rQ+1)/x), \\ X_0 &= \sum_{d \geq 2} w(d) = \sum_{r \in \mathcal{G}} \mathcal{W}(r) A_r, \\ J_0 &= \sum_{r \in \mathcal{G}} \frac{\mathcal{W}(r)}{r}. \end{aligned} \quad (4.17)$$

The equality for X_0 follows from the uniqueness in (4.16). All these sums are nonnegative.

Lemma 4.3 (Mass of the marked family). *For every fixed $\vartheta > 0$,*

$$1 \leq J_0 \ll_K 1, \quad \sum_{\substack{r \in \mathcal{G} \\ r > x^\vartheta}} \frac{\mathcal{W}(r)}{r} \ll_K \exp(-\vartheta L^{1-a_K}). \quad (4.18)$$

For $r \in \mathcal{G}$ with $r \leq x^{0.05}$, put

$$I_r = \frac{x}{4r} \int_1^2 \frac{\Psi(u)}{\log((xu-1)/(4r))} du.$$

For every fixed $A > 0$, uniformly in these r ,

$$A_r = \frac{I_r}{\varphi(M)} + O_{M, \Psi, A} \left(\frac{x}{r} L^{-A} \right), \quad I_r \asymp_\Psi \frac{x}{rL}. \quad (4.19)$$

Furthermore,

$$X_0 \asymp_{M, \Psi} \frac{x J_0}{L}. \quad (4.20)$$

The comparison constants here can be chosen independently of K and the a_i ; the threshold for x may depend on them. For every fixed $\vartheta, A > 0$,

$$\sum_{\substack{r \in \mathcal{G} \\ r > x^\vartheta}} \mathcal{W}(r) A_r = o(X_0 L^{-A}). \quad (4.21)$$

Proof. Put $s = L^{-a_K}$. Every group prime satisfies $p^s \leq e^2$, and for large x , $p^{s-1} \leq e^2/p \leq 1/2$. The bound (4.6) and an Euler product including all prime powers give

$$\sum_{r \in \mathcal{G}} \mathcal{W}(r) r^{s-1} \leq B_K \prod_{p \in \bigcup_i \mathcal{P}_i} (1 - p^{s-1})^{-1} \leq B_K \exp \left(2e^2 \sum_i V_i \right) \ll_K 1.$$

This bounds J_0 above. Restricting J_0 to $r = p_1 \cdots p_K$, with one prime $p_i \in \mathcal{P}_i$ to exponent one, contributes exactly

$$\prod_{i=1}^K \left(\frac{1}{V_i} \sum_{p_i \in \mathcal{P}_i} \frac{1}{p_i} \right) = 1.$$

The disjoint groups make these products distinct. Finally, Rankin's inequality gives

$$\sum_{\substack{r \in \mathcal{G} \\ r > x^\vartheta}} \frac{\mathcal{W}(r)}{r} \leq x^{-\vartheta s} \sum_{r \in \mathcal{G}} \mathcal{W}(r) r^{s-1} \ll_K \exp(-\vartheta L^{1-a_K}),$$

proving (4.18).

If $r \leq x^{0.05}$ and $1 \leq u \leq 2$, then, for large x ,

$$\frac{x^{0.95}}{8} \leq \frac{xu - 1}{4r} \leq \frac{x}{2}, \quad 0.9L \leq \log \left(\frac{xu - 1}{4r} \right) \leq L.$$

In particular the cutoff $Q > x^{0.9}$ is automatic on the smooth support. The prime number theorem for the fixed modulus M , uniformly over its reduced classes and with arbitrary fixed logarithmic accuracy, applied to $t \mapsto \Psi((4rt + 1)/x)$, proves the first assertion of (4.19) by partial summation. Indeed this test is supported on $t \asymp x/r$, whose logarithm is comparable to L , and its total variation is $\int |\Psi'(u)| du$, independently of r . The continuous prime main term is exactly $I_r/\varphi(M)$ after the substitution $u = (4rt + 1)/x$. The displayed logarithmic bounds and $A_\Psi > 0$ give the comparison for I_r , with constants independent of K .

Write $J_R = \sum_{r \in \mathcal{G}, r \leq x^{0.05}} \mathcal{W}(r)/r$. Taking $A = 3$ in (4.19) and summing with the marks shows that $\sum_{r \leq x^{0.05}} \mathcal{W}(r) A_r$ is bounded above and below by positive constants depending only on M, Ψ times $x J_R / L$, once x is sufficiently large. The summed error is $O_{M, \Psi}(x J_R L^{-3})$. For every r , the smooth support forces $Q < x/(2r)$, so counting all positive integers instead of primes gives $0 \leq A_r \leq x/(2r)$. Equation (4.18) therefore gives

$$J_0 - J_R \ll_K e^{-0.05 L^{1-a_K}}, \quad \sum_{\substack{r \in \mathcal{G} \\ r > x^{0.05}}} \mathcal{W}(r) A_r \ll_K x e^{-0.05 L^{1-a_K}}.$$

Since $J_0 \geq 1$, after a threshold depending on K, a_i we have $J_R \geq J_0/2$, and the last sum is at most $x J_0 / L$. Combining these bounds proves (4.20) with comparison constants depending only on M, Ψ . For general fixed $\vartheta > 0$, the same bound for A_r gives a tail $O_K(x e^{-\vartheta L^{1-a_K}})$. Divide by $X_0 L^{-A} \gg_{M, \Psi} x J_0 L^{-A-1}$ and use $J_0 \geq 1$. Since L^{1-a_K} dominates every fixed multiple of $\log L$, the quotient tends to zero. This proves (4.21). $\square$

Fix $0 < \kappa < 0.01$ and $0 < \epsilon < \kappa$, still before letting x grow. Lemma 4.3 gives $X_0 \asymp xJ_0/L$ for the total weight, with J_0 the harmonic sum that combines the estimates for fixed r . We now bound the summed divisibility remainders for $r \leq x^\epsilon$; Equation (4.21) controls the omitted tail. For odd squarefree ℓ coprime to M , define

$$\begin{aligned} D_0 &= x^{1/2-\kappa/2}, \\ A_r(\ell) &= \sum_{\substack{Q > x^{0.9} \\ Q \equiv b_r \pmod{M} \\ \ell \mid 4rQ+1}} \Psi((4rQ+1)/x), \\ g_r(\ell) &= \frac{\mathbf{1}_{\gcd(\ell, r)=1}}{\varphi(\ell)}, \quad R_r(\ell) = A_r(\ell) - g_r(\ell)A_r. \end{aligned} \tag{4.22}$$

On these squarefree integers g_r is multiplicative, and $A_r(1) = A_r$, $g_r(1) = 1$, $R_r(1) = 0$. If $\gcd(\ell, r) > 1$, both terms in $R_r(\ell)$ vanish. Otherwise the new condition prescribes the reduced class $Q \equiv -(4r)^{-1} \pmod{\ell}$. Together with $Q \equiv b_r \pmod{M}$, it gives one reduced class modulo $M\ell$.

Lemma 4.4 (Average distribution in the sieve moduli). *For every fixed $A > 0$,*

$$\sum_{\substack{r \in \mathcal{G} \\ r \leq x^\epsilon}} \mathcal{W}(r) \sum_{\substack{\ell \leq D_0 \\ \text{odd and squarefree} \\ \gcd(\ell, M)=1}} |R_r(\ell)| \ll xJ_0L^{-A}. \tag{4.23}$$

The constant and threshold may depend on the fixed parameters.

Proof. Fix $r \leq x^\epsilon$. Since $\epsilon < 0.01 < 0.05$, the cutoff in $A_r(\ell)$ is automatic on its smooth support, as in (4.19). Put $\delta_0 = (\kappa - \epsilon)/4 > 0$. For every $t \asymp x/r$ on that support, all the distinct moduli $M\ell$, $\ell \leq D_0$, satisfy

$$M\ell \leq t^{1/2-\delta_0}$$

for sufficiently large x . In fact, the exponent margin at the smallest possible scale is

$$(1 - \epsilon)(1/2 - \delta_0) - (1/2 - \kappa/2) = \frac{(\kappa - \epsilon)(1 + \epsilon)}{4} > 0,$$

which absorbs the fixed factor M and the comparison constants.

The Bombieri–Vinogradov theorem [2, 11], restricted to this fixed-power sublevel, gives for every fixed $A' > 0$ an $O(t(\log t)^{-A'})$ sum of absolute errors in prime counts up to t , using the maximum over reduced classes and main term $\text{Li}(t)/\varphi(M\ell)$. This follows from its usual level $t^{1/2}(\log t)^{-B}$, since δ_0 is fixed and positive. Integrate those errors against the derivative of $\Psi((4rt+1)/x)$. Its total absolute integral is bounded in terms of Ψ , and $t \asymp x/r$, $\log t \asymp L$ throughout. The triangle inequality under the integral thus gives

$$\sum_{\substack{\ell \leq D_0 \\ \text{odd and squarefree} \\ \gcd(\ell, M)=1}} \left| A_r(\ell) - g_r(\ell) \frac{I_r}{\varphi(M)} \right| \ll \frac{x}{r} L^{-A'}.$$

For $\gcd(\ell, r) = 1$, the main term uses $\varphi(M\ell) = \varphi(M)\varphi(\ell)$; the other terms are zero. The maximum over reduced classes in the theorem makes the bound uniform in the class depending on r, ℓ .

To replace $I_r/\varphi(M)$ by A_r , observe that

$$\sum_{\substack{\ell \leq D_0 \\ \ell \text{ squarefree}}} \frac{1}{\varphi(\ell)} \leq \prod_{p \leq D_0} \left(1 + \frac{1}{p-1} \right) = V(D_0)^{-1} \ll L.$$

Equation (4.19), with its logarithmic accuracy increased by two, absorbs this factor. Taking A' sufficiently large therefore proves $\sum_{\ell} |R_r(\ell)| \ll (x/r)L^{-A}$, uniformly in the present r . Multiplying by $\mathcal{W}(r)$, summing, and using $\sum_{r \leq x^\epsilon} \mathcal{W}(r)/r \leq J_0$ proves the lemma. $\square$

The sieve products for this progression are

$$\begin{aligned} V_M(y) &= \prod_{\substack{2 < p \leq y \\ p \nmid M}} \left(1 - \frac{1}{p-1}\right), \\ \mathfrak{S}_M &= 2 \prod_{p > 2} \left(1 - \frac{1}{(p-1)^2}\right) \prod_{p \mid M} \left(1 - \frac{1}{p-1}\right)^{-1} > 0. \end{aligned} \tag{4.24}$$

The infinite product is positive because its factors are positive and their deviations from one have a convergent sum. At every odd prime,

$$\frac{1 - 1/(p-1)}{1 - 1/p} = 1 - \frac{1}{(p-1)^2}.$$

The factor at 2 and the omitted factors at primes dividing M therefore give, with Mertens' theorem,

$$\frac{V_M(y)}{V(y)} \longrightarrow \mathfrak{S}_M, \quad V(y) \sim \frac{e^{-\gamma_E}}{\log y} \quad (y \rightarrow \infty), \tag{4.25}$$

where γ_E is Euler's constant.

Finally, for every $r \in \mathcal{G}$ with $r \leq x^{0.11}$, its distinct prime divisors exceed $\exp(L^{0.1})$. Their number is at most $0.11L^{0.9}$, and hence

$$\sum_{p \mid r} \frac{1}{p} \leq 0.11L^{0.9}e^{-L^{0.1}} = o(1). \tag{4.26}$$

For these primes, $\log(1 - 1/(p-1))^{-1} \ll 1/p$ with an absolute constant. It follows uniformly for all such r and all $y \geq 1$ that

$$\begin{aligned} \prod_{\substack{2 < p \leq y \\ p \nmid M}} (1 - g_r(p)) &= \prod_{\substack{2 < p \leq y \\ p \nmid Mr}} \left(1 - \frac{1}{p-1}\right) \\ &= V_M(y) \left(1 + O(L^{0.9}e^{-L^{0.1}})\right) = V_M(y)(1 + o(1)). \end{aligned} \tag{4.27}$$

The constant in this error can be absolute; only the threshold uses the fixed group exponents.

5 Extracting primes from the marked family

We retain the notation and weighted family of Section 4. In particular, a supported integer has the unique form $d = 4rQ + 1$, where r is a group integer with $r \leq x^{0.11}$ and $Q > x^{0.9}$ is prime, and it lies in the reduced class a modulo M . By Lemma 4.3 and $w(d) \leq B_K$, it is enough to prove that the prime weight is $\gg X_0/L$. The two sieve inputs needed below are recorded explicitly.

Input 5.1 (Block sieve). Let $z > 1$, let $\mathcal{A}$ be a finite set with nonnegative weights λ_ν for $\nu \in \mathcal{A}$, and let $\mathcal{D}$ be a set of designated primes at most z . For each $p \in \mathcal{D}$ let $\mathcal{A}_p \subseteq \mathcal{A}$

be its bad condition. For a squarefree product ℓ of primes in $\mathcal{D}$, put $\mathcal{A}_\ell = \bigcap_{p|\ell} \mathcal{A}_p$, with $\mathcal{A}_1 = \mathcal{A}$. Suppose that, including for $\ell = 1$,

$$\sum_{\nu \in \mathcal{A}_\ell} \lambda_\nu = Yg(\ell) + R(\ell), \quad Y \geq 0,$$

where g is multiplicative on these squarefree products and $g(1) = 1$. Extend $g(p)$ by zero at primes outside $\mathcal{D}$. Suppose that for fixed constants $\eta_0 > 0$ and $C_0 < \infty$,

$$0 \leq g(p) \leq 1 - \eta_0, \quad \sum_{u < p \leq u^2} g(p) \leq C_0 \quad (u > 1).$$

Write

$$S = \sum_{\nu \in \mathcal{A} \setminus \bigcup_{p \in \mathcal{D}} \mathcal{A}_p} \lambda_\nu.$$

For every sufficiently large even integer H , the weight avoiding all the bad conditions is

$$\begin{aligned} S &= Y \prod_{p \leq z} (1 - g(p)) (1 + O_{\eta_0, C_0}(e^{-H})) \\ &\quad + O_{\eta_0, C_0} \left(\sum_{\substack{\ell \leq z^{4H+2} \\ \ell \text{ squarefree} \\ p|\ell \Rightarrow p \in \mathcal{D}}} |R(\ell)| \right). \end{aligned} \tag{5.1}$$

The threshold for H and the implied constants depend only on η_0, C_0 . In particular, the constants are uniform when H grows, and the remainder sum has no additional coefficient or multiplicity factor. For $H = 2$ there is the upper bound

$$S \ll_{\eta_0, C_0} Y \prod_{p \leq z} (1 - g(p)) + \sum_{\substack{\ell \leq z^{10} \\ \ell \text{ squarefree} \\ p|\ell \Rightarrow p \in \mathcal{D}}} |R(\ell)|. \tag{5.2}$$

Both remainder sums include only squarefree products of the designated primes, with $\ell = 1$ included. This is the form stated in [P, Lemma 11.1, Equations (11.2)–(11.3)].

Input 5.2 (Rough-number density). For the density $D_\gamma(w)$ defined in Section 4, there are absolute constants $b_d, c_d, C_d > 0$ such that, for every fixed $0 < b < \min(b_d, 1/2)$,

$$\int_b^{1/2} D_t(1-t) \frac{dt}{t} = D_b(1) - 1, \quad D_b(1) \leq \frac{e^{-\gamma_E}}{b} (1 + C_d e^{-c_d/b}). \tag{5.3}$$

Here γ_E is Euler's constant, and the value of the integrand at $t = 1/2$ is immaterial. These are the assertions of [P, Lemma 11.2, Equations (11.6)–(11.7)].

The identity in (5.3) describes removal of the least factor. In the j -factor summand of $D_b(1)$, $j \geq 2$, the measure on $t_1 + \dots + t_j = 1$, $t_i \geq b$, is $\frac{1}{j!} (t_1 \dots t_j)^{-1} dt_1 \dots dt_{j-1}$. It is invariant under permutation of the coordinates. Its minimum t is unique outside a null set and is at most $1/2$. Choosing this coordinate in j ways leaves dt/t times the $(j-1)$ -factor summand of $D_t(1-t)$, since $j/j! = 1/(j-1)!$. The omitted one-factor summand of $D_b(1)$ is 1.

Measured in units of $\mathfrak{S}_M X_0/L$, the initial sieve below has main coefficient $e^{-\gamma_E}/b$, while the bins for the least prime factor cost at most $D_b(1) - 1$ apart from the partition and approximation errors. The density bound and the exponentially small loss in the initial sieve leave a positive constant for small fixed b ; a separate sieve controls the remaining balanced semiprimes.

5.1 The initial rough counts

For now take any fixed $0 < \epsilon < \kappa < 0.01$ as in Section 4, and introduce a sufficiently small fixed $0 < b < 0.01$. The choices of these parameters and of K will be made in their required order at the end. Every estimate below is for x tending to infinity after its displayed parameters have been fixed. Write

$$\Pi_r(y) = \prod_{\substack{2 < p \leq y \\ p \nmid Mr}} \left(1 - \frac{1}{p-1}\right).$$

By (4.27), $\Pi_r(y) = V_M(y)(1 + o(1))$ uniformly in y and in all supported $r \leq x^{0.11}$. A supported d is odd and is coprime to M . Consequently, for supported d , the condition $P^-(d) > z$ is equivalent to excluding divisibility by the odd primes $p \leq z$ with $p \nmid M$.

Fix a group integer $r \leq x^\epsilon$. Apply Input 5.1 to the primes Q and their weights in A_r , with the bad condition $p \mid 4rQ + 1$ at each odd $p \leq x^b$ not dividing M . The intersection counts are $A_r g_r(\ell) + R_r(\ell)$ by (4.22). Compact support of Ψ makes this a finite weighted family after removing zero weights. The densities on designated primes obey

$$0 \leq g_r(p) \leq \frac{1}{p-1} \leq \frac{1}{2}.$$

Their sums on $(u, u^2]$ have an absolute bound, since $1/(p-1) \leq 2/p$ for odd p and the prime harmonic sums on these intervals are bounded. Thus the local constants of the block sieve are absolute, independently of M, r, K .

Use the even depth

$$H_b = 2 \left\lfloor \frac{1}{40b} \right\rfloor.$$

For sufficiently small b it is an admissible depth in Input 5.1, and

$$H_b \geq \frac{1}{20b} - 2, \quad b(4H_b + 2) \leq \frac{1}{5} + 2b < 0.22 < \frac{1}{2} - \frac{\kappa}{2}. \quad (5.4)$$

In particular, every remainder modulus is at most $x^{b(4H_b+2)} \leq D_0$. Sum the resulting lower bound with the nonnegative marks $\mathcal{W}(r)$. By (4.23) the sum of the remainders is $O(xJ_0 L^{-A})$ for any fixed A ; taking $A > 2$ makes it $o(X_0/L)$ by (4.20). The tail with $r > x^\epsilon$ is negligible to this accuracy by (4.21). The uniform formula for $\Pi_r(x^b)$ and (4.25) give

$$\Pi_r(x^b) \sim V_M(x^b) \sim \frac{\mathfrak{S}_M e^{-\gamma_E}}{bL}.$$

Finally, (5.4) gives $e^{-H_b} \leq e^2 e^{-1/(20b)}$. We have proved

$$\sum_{\substack{d \geq 2 \\ P^-(d) > x^b}} w(d) \geq \frac{\mathfrak{S}_M X_0}{L} \left\{ \frac{e^{-\gamma_E}}{b} (1 - C_s e^{-c_s/b}) + o(1) \right\}, \quad (5.5)$$

where $C_s, c_s > 0$ are absolute; for example, $c_s = 1/20$ is permitted after enlarging C_s . The implied threshold may depend on all fixed parameters.

For fixed $b \leq \gamma < \gamma' \leq 1/2 - \kappa$, let

$$\mathcal{N}_{\gamma, \gamma'} = \{n \text{ prime} : x^\gamma < n \leq x^{\gamma'}\}.$$

To bound the W -rough term in the Type II replacement for $d = mn$, we use the following count conditioned on $n \mid d$:

$$\sum_{n \in \mathcal{N}_{\gamma, \gamma'}} \sum_{\substack{d \geq 2 \\ n \mid d \\ P^-(d) > W}} w(d) = X_0 V_M(W) \left\{ \log \frac{\gamma'}{\gamma} + o(1) \right\}. \quad (5.6)$$

For large x , every such n exceeds W , every prime dividing M , and every group prime. Indeed x^b eventually exceeds each of these quantities. In particular, $n \nmid r$ for every group integer r . At fixed $r \leq x^\epsilon$, condition the family of A_r on $n \mid 4rQ + 1$ and designate the odd primes $p \leq W$ not dividing M . For a squarefree product ℓ of these primes, including $\ell = 1$,

$$A_r(n\ell) = \frac{A_r}{n-1} g_r(\ell) + R_r(n\ell). \quad (5.7)$$

This follows from $n \nmid Mr\ell$ and multiplicativity of g_r . Thus the block sieve applies with $Y = A_r/(n-1)$, the same local densities, and remainder $R_r(n\ell)$. Its remainder at $\ell = 1$ is $R_r(n)$, which is explicitly allowed by Input 5.1.

Take $H_W = 2\lceil(\log L)^2\rceil$. It is an admissible growing even depth, its relative error tends to zero, and

$$W^{4H_W+2} = \exp(O(L^{0.24}(\log L)^2)) = x^{o(1)}.$$

Every remainder modulus in (5.7) therefore satisfies

$$n\ell \leq x^{1/2-\kappa+o(1)} < x^{1/2-\kappa/2} = D_0$$

for sufficiently large x . These moduli are odd squarefree and coprime to M . Moreover the map $(n, \ell) \mapsto n\ell$ is injective throughout the sum: n is the unique prime factor of this product exceeding W . Hence the unweighted remainder sums of the block sieve, summed over n and then with the marks over r , form a subsum of (4.23), without any multiplicity or depth factor. Taking its accuracy $A > 2$ makes this $o(X_0 V_M(W))$, because $X_0 V_M(W) \asymp_{M, \Psi} x J_0 L^{-1.24}$.

The main sieve product is $\Pi_r(W) = V_M(W)(1 + o(1))$ uniformly. Also the prime harmonic estimate gives

$$\sum_{n \in \mathcal{N}_{\gamma, \gamma'}} \frac{1}{n-1} = \log \frac{\gamma'}{\gamma} + o(1).$$

Here replacing $1/n$ by $1/(n-1)$ costs $o(1)$, and Mertens' theorem evaluates the sum of $1/n$. These facts give the main term in (5.6) for $r \leq x^\epsilon$. For the omitted tail, a supported $d < 2x$ has at most $\log(2x)/(bL) = O_b(1)$ distinct prime divisors above x^b . Multiplying the tail bound (4.21) by this fixed factor still gives $o(X_0 V_M(W))$. This proves (5.6).

5.2 Removing the least prime factor

If the least prime factor of a supported composite d belongs to the bin $(x^\gamma, x^{\gamma'}]$, write it as n and put $m = d/n$. Then $P^-(m) \geq n > x^\gamma$, also when $n^2 \mid d$. The weight of these composites is consequently at most

$$T_{\gamma, \gamma'} = \sum_{n \in \mathcal{N}_{\gamma, \gamma'}} \sum_{\substack{m \geq 1 \\ P^-(m) > x^\gamma}} w(mn).$$

Lemma 4.2 gives the replacement

$$T_{\gamma, \gamma'} = \sum_{n \in \mathcal{N}_{\gamma, \gamma'}} \sum_{\substack{m \geq 1 \\ P^-(m) > W \\ x < mn < 2x}} \frac{D_\gamma(\log m/L)}{LV(W)} w(mn) + o(X_0/L). \quad (5.8)$$

We check the geometry, support, and total error in this application.

Partition each factor into disjoint half-open dyads $[H_m, 2H_m)$ and $[H_n, 2H_n)$, restricting n to its bin. Retain every rectangle containing a real point with $mn/x \in \text{supp } \Psi$ and $x^\gamma < n \leq x^{\gamma'}$. In such a rectangle $x/4 < H_m H_n < 2x$. There are $O(L)$ rectangles: there are $O(L)$ possible dyads for n , and the displayed product constraint allows only an absolute bounded number of dyads for m for each of them. On the support of $\Psi(mn/x)$, write $t = \log n/L$. Then

$$\frac{\log m}{L} = 1 - t + O(L^{-1}), \quad b \leq \gamma < t \leq \gamma' \leq \frac{1}{2} - \kappa. \quad (5.9)$$

Passing from a point in this support to the whole m -dyad changes its logarithmic exponent by at most $\log 2/L$. Therefore, for all large x , the full m -dyad is in the exponent interval

$$[w_-, w_+] = \left[\frac{1}{2} + \frac{\kappa}{2}, 1 - \frac{b}{2} \right].$$

In particular $0 < \gamma < w_- < w_+ < 1$. Both dyad scales exceed $x^{b/2}$ for large x : the n -scale is at least $x^{b/2}$, while the m -scale is at least $x^{1/2+\kappa/2}$.

For the separation of the smooth weight, put

$$\hat{\psi}(v) = \frac{1}{2\pi} \int_{\mathbb{R}} \Psi(e^s) e^{-ivs} ds.$$

Fourier inversion and smooth compact support give

$$\begin{aligned} \Psi(mn/x) &= \int_{\mathbb{R}} \hat{\psi}(v) x^{-iv} m^{iv} n^{iv} dv, \\ \int_{\mathbb{R}} |\hat{\psi}(v)| dv &\ll_{\Psi} 1, \quad \int_{|v| > L} |\hat{\psi}(v)| dv \ll_{A, \Psi} L^{-A} \end{aligned} \quad (5.10)$$

for every fixed $A > 0$. Insert the character expansion (4.14) to separate the progression condition.

In each retained box apply Lemma 4.2 with

$$\delta = b/2, \quad C = 2, \quad D_* = 6,$$

and with the exponent interval above. For $|v| \leq L$ its first coefficient is the difference of the two tests in (5.8), multiplied by $\chi_0(m)m^{iv}$. The second coefficient is $\chi_0(n)n^{iv}\mathbf{1}_{n \text{ prime}}$, restricted to its bin and dyad. It is bounded by one and is supported on W -rough integers, since $n > W$. The function F is exactly the invariant function in (4.15). The product condition remains in Ψ , so neither coefficient has been restricted by a condition on both factors. The lemma applies for sufficiently large K depending on these fixed parameters, independently of the band exponents. Its bound (4.13) is $O(xL^{-6})$ per box; integration over $|v| \leq L$ costs only the bounded L^1 norm in (5.10).

For $|v| > L$, the density is bounded on the fixed exponent interval and $LV(W) \asymp L^{0.76}$. Thus the two discrepancy coefficients are bounded for large x . The marked weight is

bounded by B_K by (4.6), and a box has $O(H_m H_n) = O(x)$ integer pairs. The last bound in (5.10), with sufficiently large fixed A , makes this tail $O_{K,\Psi}(xL^{-6})$ per box as well. The $O(L)$ boxes therefore contribute $O(xL^{-5}) = o(X_0/L)$, with the constant allowed to depend on all fixed parameters, using (4.20) and $J_0 \geq 1$. This proves (5.8). All occurrences of the density in that equation are in its domain, by (5.9).

By the joint continuity established in Section 4, (5.9) implies, uniformly on the support in (5.8),

$$D_\gamma(\log m/L) \leq \sup_{\gamma \leq t \leq \gamma'} D_\gamma(1-t) + o(1).$$

Since $n > W$, the condition $P^-(m) > W$ is equivalent to $P^-(mn) > W$. For a fixed divisor $n \mid d$ the factor $m = d/n$ is unique. We may therefore apply (5.6) after changing variables $d = mn$. Together with $V_M(W)/V(W) \rightarrow \mathfrak{S}_M$, this gives

$$T_{\gamma,\gamma'} \leq \frac{\mathfrak{S}_M X_0}{L} \left\{ \left(\sup_{\gamma \leq t \leq \gamma'} D_\gamma(1-t) \right) \log \frac{\gamma'}{\gamma} + o(1) \right\}. \quad (5.11)$$

Choose a finite partition $b = \gamma_0 < \gamma_1 < \dots < \gamma_J = 1/2 - \kappa$. The set of pairs $(\gamma, 1-t)$ with $b \leq \gamma \leq t \leq 1/2 - \kappa$ is a compact subset of $\{(\gamma, w) : w > \gamma > 0\}$, since $1-t-\gamma \geq 2\kappa$. Joint continuity of the density is uniform there. For every t in a bin, the bin supremum in (5.12), whose first argument is fixed at the left endpoint, differs from $D_t(1-t)$ by a quantity tending uniformly to zero with the mesh. The measure dt/t has finite total mass $\log((1/2 - \kappa)/b)$. We may consequently choose a sufficiently fine fixed partition for which

$$\begin{aligned} \sum_{j=1}^J \left(\sup_{\gamma_{j-1} \leq t \leq \gamma_j} D_{\gamma_{j-1}}(1-t) \right) \log \frac{\gamma_j}{\gamma_{j-1}} &\leq \int_b^{1/2-\kappa} D_t(1-t) \frac{dt}{t} + \frac{1}{8} \\ &\leq D_b(1) - 1 + \frac{1}{8}. \end{aligned} \quad (5.12)$$

The last inequality uses nonnegativity of D_t and Input 5.2.

Put

$$U_\kappa = \sum_{\substack{d \geq 2 \\ P^-(d) > x^{1/2-\kappa}}} w(d).$$

Every supported integer counted in (5.5) but not in the sum defining U_κ is composite: it exceeds x , whereas its least prime factor is at most $x^{1/2-\kappa}$. That factor lies in one of the bins of our partition. Subtracting (5.11) for these finitely many bins from (5.5), and using (5.12) and (5.3), yields

$$\frac{LU_\kappa}{\mathfrak{S}_M X_0} \geq 1 - \frac{1}{8} - \frac{e^{-\gamma_E}}{b} \left(C_s e^{-c_s/b} + C_d e^{-c_d/b} \right) + o(1). \quad (5.13)$$

Here the lower bound uses $\frac{e^{-\gamma_E}}{b}(1 - C_s e^{-c_s/b}) - (D_b(1) - 1 + 1/8)$; the stated upper bound on $D_b(1)$ gives precisely the two exponential losses in (5.13).

5.3 Balanced composites and the parameter choices

It remains to bound the composite part of U_κ . The final choice of κ precedes those of b and K , so the coefficient in this bound must be independent of all three. We achieve this by sieving pairs for each fixed r before summing the marks. Such a composite has exactly two prime factors counted with multiplicity. Indeed three factors, each greater than $x^{1/2-\kappa}$,

would have product at least $x^{3/2-3\kappa} > 2x$ for large x . Write the two factors as m, n . Each is greater than $x^{1/2-\kappa}$ and less than $2x^{1/2+\kappa}$, so for large x both belong to

$$[x^{1/2-2\kappa}, x^{1/2+2\kappa}].$$

Use ordered pairs (m, n) to bound their weight; this counts a square once and may count other semiprimes twice, which is harmless for an upper bound.

Cover these pairs by disjoint dyadic boxes $[M_1, 2M_1) \times [M_2, 2M_2)$ meeting the displayed factor range and $x < mn < 2x$. Every retained box satisfies

$$\frac{x}{4} < M_1 M_2 < 2x, \quad M_i \leq 2x^{0.54} \quad (i = 1, 2). \quad (5.14)$$

The latter is a uniform relaxed bound because $\kappa < 0.01$. There are at most $C_{\text{dyad}}(\kappa L + 1)$ boxes, for an absolute constant C_{dyad} . To see this, the logarithmic length of the factor range is $4\kappa L$, so it meets $O(\kappa L + 1)$ dyads for m ; for each of them the first inequality in (5.14) permits only an absolute bounded number of dyads for n .

Fix one box and a group integer r in the full support, so $r \leq x^{0.11}$. Put $k_r = 4r$ and $z = x^{0.002}$. A prime pair contributing to the weight through this r obeys

$$mn \equiv 1 \pmod{k_r}, \quad mn \not\equiv 0, 1 \pmod{j} \quad (j \leq z \text{ an odd prime, } j \nmid r). \quad (5.15)$$

In fact $m, n > x^{1/2-2\kappa} > z$ and $Q = (mn - 1)/(4r) > x^{0.9} > z$ are primes. Thus $mn \equiv 0 \pmod{j}$ would force j to be one of the prime factors m, n , and, since $j \nmid 4r$, $mn \equiv 1 \pmod{j}$ would force $j = Q$. Both are impossible. We discard the progression condition modulo M to obtain an upper bound.

Use as a finite unweighted family all integer pairs in the box satisfying the first congruence in (5.15), and designate its odd primes $j \leq z$ not dividing r , with bad condition $mn \equiv 0$ or $1 \pmod{j}$. All sums and products indexed by j in this balanced sieve are over primes. For a squarefree product ℓ of these primes set

$$\rho(\ell) = \prod_{j|\ell} (3j - 2), \quad \rho(1) = 1.$$

There are $\varphi(k_r)$ residue pairs modulo k_r with product one. At j there are $2j - 1$ residue pairs with product zero and $j - 1$ with product one, and these two sets are disjoint. Since $(k_r, \ell) = 1$, the Chinese remainder theorem gives exactly $\varphi(k_r)\rho(\ell)$ residue pairs modulo $k_r\ell$ satisfying the base congruence and all the designated bad conditions at primes dividing ℓ .

Each residue pair modulo $k_r\ell$ occurs in the box

$$\frac{M_1 M_2}{(k_r \ell)^2} + O\left(\frac{M_1 + M_2}{k_r \ell} + 1\right)$$

times. Thus its intersection count has the form $Y_r g(\ell) + R(\ell)$, where

$$Y_r = \frac{M_1 M_2 \varphi(k_r)}{k_r^2}, \quad g(\ell) = \frac{\rho(\ell)}{\ell^2}. \quad (5.16)$$

$$\begin{aligned} |R(\ell)| &\ll \varphi(k_r) \rho(\ell) \left(\frac{M_1 + M_2}{k_r \ell} + 1 \right) \\ &\ll (M_1 + M_2) \ell + k_r \ell^2. \end{aligned} \quad (5.17)$$

The last inequality uses $\varphi(k_r) \leq k_r$ and $\rho(\ell) \leq \ell^2$, because $3j - 2 \leq j^2$ for every odd prime j . This count also includes $\ell = 1$, so its boundary error for the base family is included in $R(1)$.

For a designated prime, $g(j) = (3j - 2)/j^2 \leq 7/9$ and $g(j) \leq 3/j$. Thus the local gap and block sums in Input 5.1 again have absolute constants. Its $H = 2$ upper bound uses only $\ell \leq z^{10} = x^{0.02}$. Summing (5.17) over this range, bounding the designated squarefree products by all positive integers, gives

$$\begin{aligned} \sum_{\ell \leq x^{0.02}} |R(\ell)| &\ll (M_1 + M_2) \sum_{\ell \leq x^{0.02}} \ell + k_r \sum_{\ell \leq x^{0.02}} \ell^2 \\ &\ll x^{0.54+0.04} + rx^{0.06} \ll x^{0.58} + x^{0.17}. \end{aligned} \quad (5.18)$$

The notation in the left-hand sum restricts to the designated products for which R is defined. The exponent margins compared with x/r are uniform throughout $r \leq x^{0.11}$:

$$\frac{x^{0.58}}{x/r} = rx^{-0.42} \leq x^{-0.31}, \quad \frac{rx^{0.06}}{x/r} = r^2x^{-0.94} \leq x^{-0.72}.$$

Both fixed power savings absorb L^4 . It follows that (5.18) is $o((x/r)L^{-4})$ uniformly in every supported r , with no dependence on K in this assertion.

For the main product the exact local identity is

$$1 - g(j) = \left(1 - \frac{1}{j}\right)^3 \left(1 - \frac{1}{(j-1)^2}\right).$$

The last factor is at most one. The missing factor at two costs only the factor 8 when compared with $V(z)^3$. The missing factors at primes dividing r cost at most

$$\prod_{j|r} \left(1 - \frac{1}{j}\right)^{-3} = \exp \left(O \left(\sum_{j|r} \frac{1}{j} \right) \right) = 1 + o(1)$$

uniformly, since on the full support $\sum_{j|r} j^{-1} \leq 0.11L^{0.9}e^{-L^{0.1}}$ by (4.26). Consequently

$$\prod_{\substack{j \leq z \\ j \text{ odd prime} \\ j \nmid r}} (1 - g(j)) \ll V(z)^3 \ll L^{-3}, \quad (5.19)$$

with an absolute constant. The numerical exponent 0.002 is fixed, so the last estimate follows from Mertens' theorem with a fixed constant.

By (5.14), $Y_r \ll x/r$. Combining (5.2), (5.18), and (5.19) bounds the number of prime pairs in this box contributing through r by

$$O((x/r)L^{-3}),$$

uniformly for all supported r . Its constant is absolute; the power-saving error above has been absorbed for sufficiently large x . The smooth weight is at most one. Multiplying by $\mathcal{W}(r)$ and summing over the full support of r , without discarding a tail, therefore bounds the weighted prime pairs in one box by $C_{\text{pair}}xJ_0/L^3$ with an absolute C_{pair} . The unique representation of the support justifies this assignment of each weighted pair to its group integer.

Let $\mathcal{C}_\kappa$ denote the weight of the composites counted by U_κ . Summing the last estimate over the boxes gives

$$\mathcal{C}_\kappa \leq C_2(\kappa + L^{-1}) \frac{X_0}{L}. \quad (5.20)$$

Here C_2 depends only on M, Ψ and the fixed dyadic comparison conventions, and is independent of κ, b, K . Indeed the preceding bound before converting the normalization is $C_{\text{pair}} C_{\text{dyad}}(\kappa L + 1)xJ_0/L^3$. The lower comparison in (4.20) is $X_0 \geq c_{M,\Psi} xJ_0/L$ for large x , with $c_{M,\Psi} > 0$ independent of K . We may take $C_2 = C_{\text{pair}} C_{\text{dyad}}/c_{M,\Psi}$. The threshold for x may depend on all the fixed parameters; this does not change the stated independence of the constant.

Every supported prime is counted by U_κ . Subtracting its composite part using (5.13) and (5.20) yields

$$\begin{aligned} \frac{L}{\mathfrak{S}_M X_0} \sum_{d \text{ prime}} w(d) &\geq 1 - \frac{1}{8} - \frac{e^{-\gamma_E}}{b} \left(C_s e^{-c_s/b} + C_d e^{-c_d/b} \right) \\ &\quad - \frac{C_2}{\mathfrak{S}_M} (\kappa + L^{-1}) + o(1). \end{aligned} \quad (5.21)$$

We now choose the parameters. With M fixed, the function Ψ and the dyadic convention have already fixed C_2 , while $\mathfrak{S}_M > 0$ depends only on M . Choose $0 < \kappa < 0.01$ so small that $C_2 \kappa / \mathfrak{S}_M < 1/8$, and then choose $0 < \epsilon < \kappa$. Next choose fixed $b > 0$ sufficiently small for Inputs 5.1 and 5.2, for (5.4), and so that

$$\frac{e^{-\gamma_E}}{b} \left(C_s e^{-c_s/b} + C_d e^{-c_d/b} \right) < \frac{1}{8}.$$

This is possible because $b^{-1}e^{-c/b} \rightarrow 0$ as $b \downarrow 0$ for each $c > 0$. Fix a finite partition satisfying (5.12). For its finitely many left endpoints γ , use the fixed Type II choices $\delta = b/2$, $C = 2$, $D_* = 6$ and the exponent interval already specified. Now choose K sufficiently large for Lemma 4.2 in every one of these applications. The lower bound on K is independent of the band exponents, so choose any fixed $0.1 < a_1 < \dots < a_K < 0.2$ afterwards. Finally let x exceed all thresholds for these fixed choices.

The three fixed losses in (5.21) are then at most $1/8$ each, and the term $C_2/(\mathfrak{S}_M L)$ and the $o(1)$ term tend to zero. In particular, for all sufficiently large x ,

$$\sum_{d \text{ prime}} w(d) \geq \frac{\mathfrak{S}_M X_0}{2L} \gg_{M,\Psi} \frac{xJ_0}{L^2} \gg_{M,\Psi} \frac{x}{L^2}.$$

By (4.16), each prime with positive weight has the required support and progression properties, and its weight is at most B_K . Dividing the prime mass by this fixed bound gives the required lower bound for distinct primes. This proves Proposition 2.1.

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